



Device modelling and numerical analysis of high-efficiency double absorbers solar cells with diverse transport layer materials

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ABSTRACT

Recently, a lot of focus has been placed on cesium lead iodide (CsPbI₃) since it is considered a potentially useful inorganic halide perovskite with improved stability. Moreover, another potential absorber material is the mixed chalcogenide CZTSSe, which is abundant on Earth, cheap, environmentally acceptable, and has excellent photovoltaic performance. This research numerically simulated a novel double absorber solar cell structure employing CsPbI₃ and CZTSSe absorbers in the active layer in SCAPS-1D. The current study analyses the effects of various electron and hole transport materials, back contact material's work functions, working temperatures, variations in defect concentration, and absorber thickness on the performance of photovoltaic devices. After researching a variety of distinct arrangements of double absorber solar cells, it was realized that the FTO/STO/CsPbI₃/CZTSSe/NiO/W cell configuration exhibited the best overall performance with an open circuit voltage (V_{oc}) at 1.0207 V, a short circuit current density (J_{sc}) at 41.815426 mA/cm², Fill Factor (FF) at 87.50 %, and a Power Conversion Efficiency (PCE) at 37.35 %. The modeling of the device showed that a thickness of around 1.4 μ m for the CZTSSe absorber is optimal. This simulation shows that when the working temperature in the cell and the defect concentration in the absorber increase, the efficiency of the device reduces uniformly, and the device is stable at 300 K temperature. In conclusion, if the material's work function is greater than 5.20 eV, then it is suitable for use as an anode.

1. Introduction

Solar energy and other renewable technologies are widely seen as viable options for clean energy sources that may alleviate the negative environmental consequences of the coal and power sectors, minimize ancillary waste, and provide a long-term answer to the world's energy issues. Single-junction solar cells cannot overcome their Shockley-Queasier limit because of the absorber material's bandgap that limits the quantity of light that can be absorbed (E_g). Besides, researchers are very interested in organic-inorganic metal halide materials for photovoltaics because of their unique properties such as larger absorption coefficient, greater charge carriers' mobility, extended electron and hole diffusion length, adjustable bandgap creation, large dielectric constant, solution-treated manufacturing process (Li et al., 2020). More recently, double absorber solar cells have been extensively researched to improve device performance further because a solar cell with two absorber layers will be more effective since it can collect solar photons throughout a wider spectrum. In this context, Maklavani et al. realize a CZTSSe/CZTS-

based solar cell by using a solution processing method, and the device's overall efficiency comes out to 17.81 % (Enayati Maklavani and Mohammadnejad, 2020). However, double-absorber solar cells were successfully constructed by Alzoubi et al., achieving a PCE of 19.40 % (Alzoubi et al., 2021). High-efficiency perovskite/perovskite double absorber solar cells were investigated by Abedini-Ahangarkola et al. utilizing SCAPS-1D, with an overall Efficiency of 30.29 % (Abedini-Ahangarkola et al., 2022).

Double absorber solar cells have been the subject of significant research and development. Yet, single-layer perovskite cells are more efficient than these devices, but they still have room to improve. As a result, academics put significant efforts into designing Double-absorber solar cells. In this context, Inorganic halide perovskite materials CsPbX₃ (where, X = I, Br, Cl) have been reported to have greater thermostability than that of the organic-inorganic hybrid perovskite (Sutton et al., 2016) and noteworthy PV performance, in which CsPbI₃ with cubic structure shows the optimal bandgap of 1.73 eV for PV cells (Eperon et al., 2014; Chen et al., 2019), and is also a suitable candidate to design

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double absorber cells (Ahmad et al., 2017). However, CZTSSe is an excellent material for PV applications (Simya et al., 2016; Moham-madnejad and Baghban Parashkoush, 2017) because of its kesterite-type crystalline form as a stable crystalline stage, components with no toxicity, the composition of Earth-abundancy, cost-effective power generation (Khattak et al., 2018), higher adsorption coefficient $>10^4 \text{ cm}^{-1}$, and the tunability of the band gap, which varies wildly from 0.95 eV to 1.5 eV. Besides, Choosing the appropriate ETM and HTM of this structure can enhance performance, boost hole extraction from the perovskite, and make it easier to move through structures, all of which contribute to improving the repeatability and stability of double absorber solar cells. To maintain consistency and the appropriate qualities, the ETL and HTL layers should be deposited utilizing techniques such solution processing, chemical vapor deposition, and physical vapor deposition. Spin-coating, that's preferred method for the absorber layer is a popular approach for depositing perovskite films because it is easy to use, can be scaled up, and produces high-quality films.

Here, SCAPS-1D was utilized to design and simulate a hybrid double-absorber solar cell manufactured by CsPbI_3 and CZTSSe as the top and bottom absorbers. The motivation underlying this combination is to improve the overall stability and performance of the solar cell by taking use of the complementing qualities of both materials. In order to maximize light absorption and charge carrier production by utilizing both materials, it is intended to alleviate stability difficulties frequently associated with pure perovskite-based devices by integrating CZTSSe into the device design alongside perovskite. Furthermore, the fusion of CZTSSe with perovskite enables the investigation of innovative device topologies and the enhancement of device efficiency. Because of the limitations of SCAPS-1D, it is considered that there is a precise tunneling link between the top and bottom absorbers free of optical and electrical loss. Double absorber solar cells' performance is studied using various transport materials, including PCBM, STO, C_{60} , CdS as ETL and NiO, Cu_2Te , CuSCN, and spiro-OMeTAD as HTL. Additionally, the influence of operating temperature, defect concentration of the active layer, and absorber layer thickness on the efficiency of the proposed double active layer solar cell is explored. Device combinations with promising performance include solar cells with several different back contacts like carbon (5.0 eV), gold (5.1 eV), tungsten (5.2 eV), Palladium (5.3 eV), nickel (5.5 eV), platinum (5.65 eV), and selenium (5.9 eV), are studied accordingly (Tariq Jan and Noman, 2022; Omarova et al., 2022; Deepthi Jayan and Sebastian, 2021; Raoui et al., 2019). It is found that the optimal double absorber structure for absorbers having an energy gap of 1.73 and 1.15 eV achieved a PCE of 37.35 %. The findings of our proposed structure will provide researchers working on future experiments with a plethora of data.

2. Simulation methodology and device structure

2.1. Simulation methodology

The simulations were carried out with the most recent version, 3.3.10, of the SCAPS-1D numerical program. This 1-dimensional solar cell modelling program was created at Gent University in Belgium (Burgelman et al., 2013). It is chosen to utilize SCAPS as our solar device simulator because the findings of its simulations are in excellent agreement with those obtained from previous experiments (Sharbati et al., 2018) and are preferable to other solar device simulator software. Fig. 1 depicts the flowchart of the simulation process for SCAPS-1D. The structure can add as many as seven distinct layers when using SCAPS. After the structure is specified, solving the following differential equations allows SCAPS to be utilized to compute a variety of PV performance metrics:

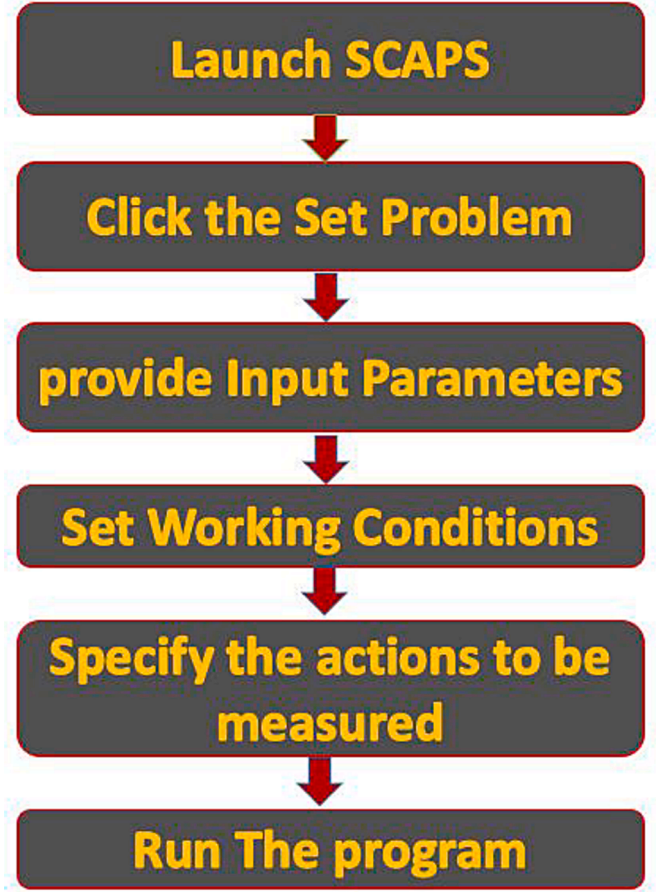


Fig. 1. Flow chart of Simulation Procedure for SCAPS-1D.

$$\text{Poisson Equations'}: \frac{\partial^2 \phi}{\partial x^2} = -\frac{\partial E}{\partial x} = -\frac{\rho}{\epsilon_s} = -\frac{q}{\epsilon_s} [p - n + N_D(x) - N_A(x) \pm N_{def}(x)] \quad (1)$$

$$\text{Transport Equation: } J_{n,p} = nq\mu_n E + qDn \frac{\partial n}{\partial x} + pq\mu_p E + qDp \frac{\partial p}{\partial x} \quad (2)$$

$$\text{Diffusion length: } L_{n,p} = \sqrt{D_{n,p} \tau_{n,p}} \quad (3)$$

$$\text{Diffusivity } D_{n,p} = \left[\left(\frac{k_B T}{q} \right) \mu_{n,p} \right] \quad (4)$$

$$\text{Continuity Equation: } \frac{\partial n,p}{\partial t} = \frac{1}{q} \frac{\partial J_{n,p}}{\partial x} + (G_n - R_n) + \frac{1}{q} \frac{\partial J_{p,p}}{\partial x} + (G_p - R_p). \quad (5)$$

$$\text{Open circuit voltage: } v_{oc} = \frac{nk_B T}{q} \left[\ln \left(\frac{I_L}{I_0} + 1 \right) \right] \quad (6)$$

$$\text{Carrier lifetime: } \tau = \frac{1}{\sigma N_i v_{th}} \quad (7)$$

Where ϕ = cells electrostatic potential, E is the electric field, ϵ = materials permittivity, n & p = free electron and hole concentration, N_A & N_D are as doping concentration, N_{def} = probable defect concentration, $\mu_{n,p}$ = mobility of electron or hole, ρ and q are referred to as elemental charges $\tau_{n,p}$ = lifetime of electron or hole, $\frac{\partial n,p}{\partial x}$ = concentration gradient of electron or hole, $G_{n,p}$ = rate of optical generation, $R_{n,p}$ = rate of recombination, $\frac{k_B T}{q}$ = thermal voltage, I_L = generated current by light

and I_0 = saturation current (Tariq Jan and Noman, 2022).

2.2. Device structure

The proposed design of a Double absorber solar cell is FTO/STO/CsPbI₃/CZTSSe/NiO/W, where FTO is utilized as a transparent conducting oxide (TCO), STO as ETL, CsPbI₃ & CZTSSe as active layers, NiO as HTL, and Tungsten (W) as a back electrode are depicted in Fig. 2(a). Here, all simulations have been performed at 300 K, with a frequency of 1.0×10^6 Hz and illumination at the usual AM 1.5 G 1 sun. Firstly, simulation was used to optimize the absorber layer's thickness, and the other layers' thicknesses were set according to the literature. In this case, a tunnel connection between two absorbers with zero electrical resistance and zero optical loss is imagined. Double absorber solar cells should include a conformal recombination layer inserted between the absorber layers to reduce shunting and shield the top layers from solvents and sputtering damage (Abedini-Ahangarkola et al., 2022). An illustration of the proposed structure's energy levels is depicted in Fig. 2 (b). Fig. 2(c) depicts the external quantum efficiency (EQE) spectra of optimized device structure, where the wavelengths changed from 200 to 1000 nm. Parameters for the electrical systems employed in the simulations are summarized in Table 1; these values were derived from published experimental and computational work, where E_g represents the bandgap, W represents the thickness of the layer, ϵ_r represents the relative dielectric constant, N_v , and N_c represents the effective density of states for valance band and conduction band accordingly, χ represents

the electron affinity, v_{te} , v_{th} , μ_n , and μ_p indicate the thermal velocity and electrons and holes mobility, N_D and N_A stands for donors and acceptors density and N_t represents the density of total defects. In addition, Table 2 also specifies the values for the parameters that control the relative strengths of the ETL/CsPbI₃ and CZTSSe/HTL interactions. Here, the total interface defect density was kept constant throughout the experiment to investigate the impact of defect density. In addition, the interface characteristics shared by ETL/CsPbI₃ and CZTSSe/HTL are detailed in Table 2.

3. Results and discussions

3.1. Influence of several ETLs depending on NiO as HTL on Double absorber solar cell

This section explores various substances as ETL, with NiO serving as the HTL. Since the ETL layer is essential for effective electron extraction to the front contact from the absorber layer, so it is crucial to choose an appropriate ETL material to maximize performance. Table 3 summarizes the literature-referenced simulation parameters for evaluating the influence of distinct ETL materials on device functionality. Several types of ETL materials, including PCBM, STO, C₆₀, and CdS, are investigated. Band alignment diagrams for several ETLs with top absorber CsPbI₃ are shown in Fig. 3. Table 4 summarizes the results of a comparison of the Performance of various ETLs used in Double absorber structures, revealing that the structure using STO as ETL achieves the highest

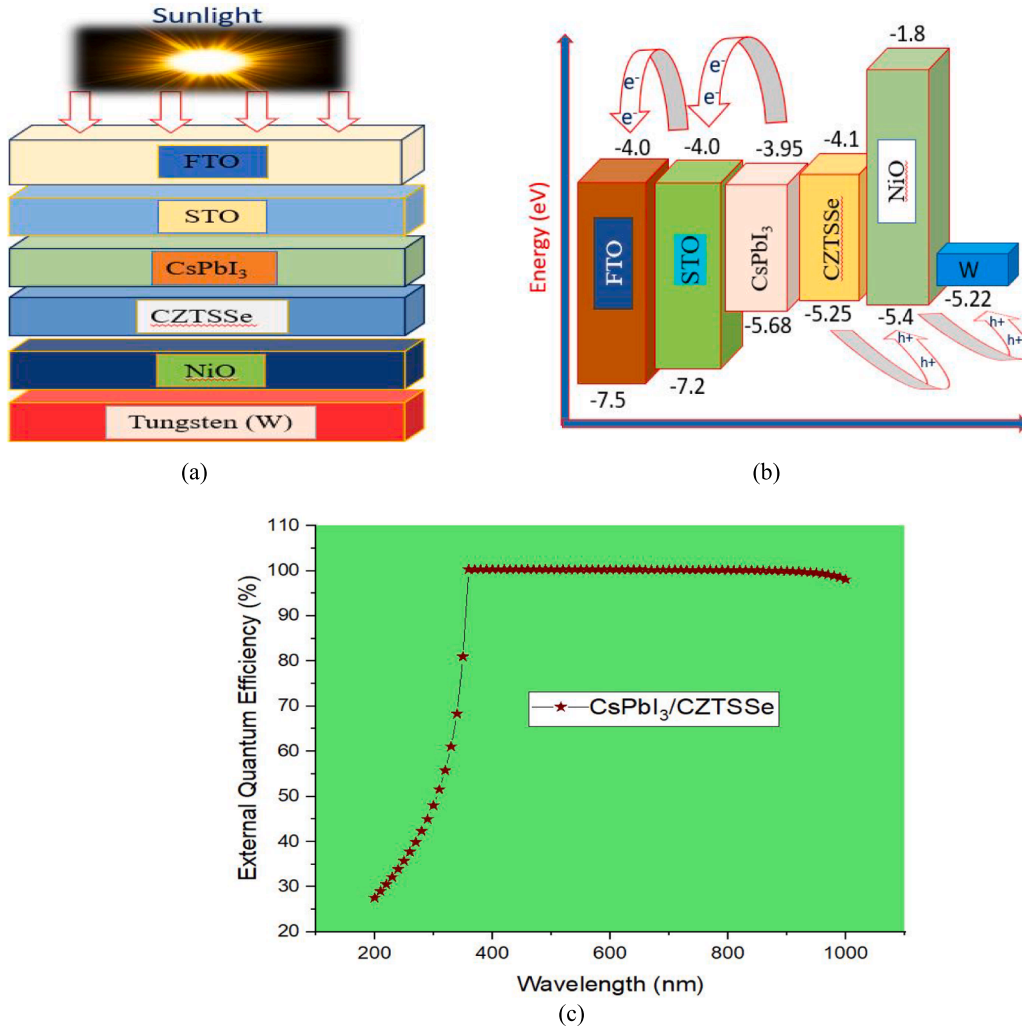


Fig. 2. Schematic of Proposed: (a) device structure, (b) Energy Band Alignment Diagram, (c) external quantum efficiency spectra.

Table 1

Parameters that are utilized to conduct this simulation:

Parameters	FTO (Rai et al., 2020)	STO (Singh et al., 2020)	CsPbI ₃ (Lin et al., 2020)	CZTSSe (Et-taya et al., 2020)	NiO (Abedini-Ahangarkola et al., 2022)
W (μm)	0.4	0.07	0.01	1.4	0.350
E _g (eV)	3.5	3.2	1.73	1.15	3.6
x (eV)	4.00	4.00	3.95	4.1	1.8
C _r	9.0	8.7	6.0	13.60	11.75
N _c (1/cm ³)	2.02×10^{18}	1.7×10^{19}	1.1×10^{20}	2.2×10^{18}	2.5×10^{20}
N _v (1/cm ³)	1.80×10^{19}	2×10^{20}	8×10^{19}	1.8×10^{19}	2.5×10^{20}
V _{te} (cm ² /Vs)	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7
V _{tp} (cm ² /Vs)	1×10^7	1×10^7	1×10^7	1×10^7	1×10^7
μ _n (cm ² /Vs)	20	5.3×10^3	16	100	2×10^{-1}
μ _p (cm ² /Vs)	10	6.6×10^2	16	25	2×10^{-1}
N _D (1/cm ³)	2×10^{19}	2×10^{16}	0	0	0
N _A (1/cm ³)	0	0	1×10^{15}	1×10^{18}	5×10^{18}
N _t (1/cm ³)	1×10^{15}	1×10^{15}	2.07×10^{14}	1×10^{12}	1×10^{15}

Table 2

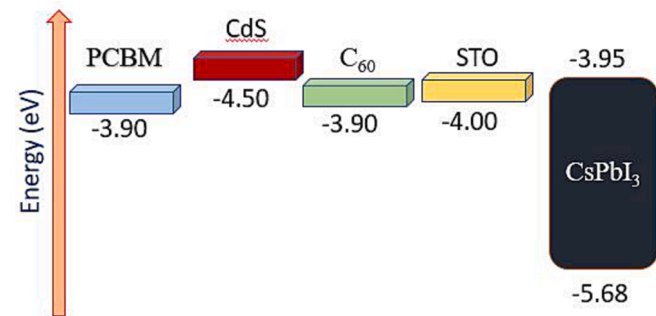
Interface Defect Parameters used in this simulation.

Parameters	ETM/CsPbI ₃	CZTSSe/HTM
Defect Type	Neutral	Neutral
A _e (cm ²) (Cross Section area of electrons)	1.0×10^{-19}	1.0×10^{-19}
A _h (cm ²) (Cross Section area of holes)	1.0×10^{-19}	1.0×10^{-19}
Energetic Distribution	Single	Single
Reference for defect energy level E _t	Above the highest E _v	Above the highest E _v
Energy with respect to a reference (eV)	0.600	0.600
Total density (cm ⁻³)	1×10^{10}	1×10^{10}

Table 3

The numerical parameters that were taken for each ETL Material.

Parameters	STO (Singh et al., 2020)	PCBM (Abedini-Ahangarkola et al., 2022)	C ₆₀ (Lakhdar and Hima, 2020)	CdS (Chowdhury et al., 2020)
W (μm)	0.07	0.2	0.05	0.045
E _g (eV)	3.2	2.0	1.7	2.4
x (eV)	4.00	3.90	3.9	4.5
C _r	8.7	3.90	4.2	10.0
N _c (1/cm ³)	1.7×10^{19}	2.5×10^{21}	8×10^{19}	2.2×10^{18}
N _v (1/cm ³)	2×10^{20}	2.5×10^{21}	8×10^{19}	1.9×10^{19}
V _{te} (cm ² /Vs)	1×10^7	1×10^7	1×10^7	1×10^7
V _{tp} (cm ² /Vs)	1×10^7	1×10^7	1×10^7	1×10^7
μ _n (cm ² /Vs)	5.3×10^3	2×10^{-2}	8×10^{-2}	3.5×10^2
μ _p (cm ² /Vs)	6.6×10^2	2×10^{-2}	3.5×10^{-3}	2.5×10^1
N _D (cm ⁻³)	2×10^{16}	1×10^{19}	2.6×10^{18}	1×10^{18}
N _A (cm ⁻³)	0	0	0	0
N _t (cm ⁻³)	1×10^{15}	1×10^{15}	1×10^{14}	1×10^{14}

**Fig. 3.** Band Alignment of Top Absorber with a Variety of Electron Transfer Layers.**Table 4**

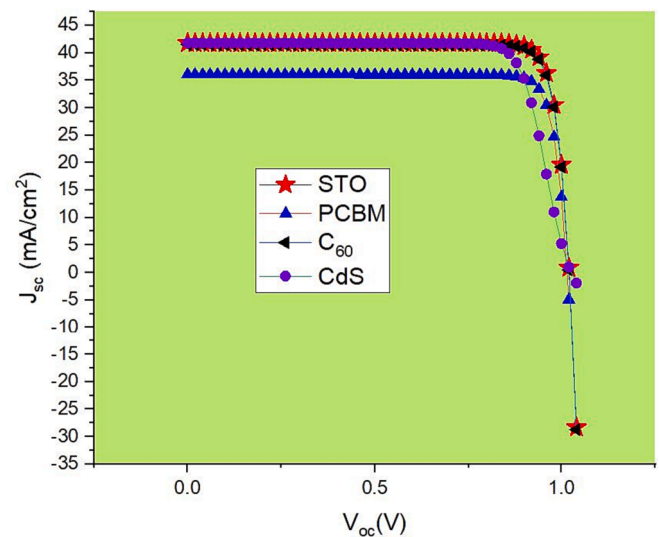
A comparison of the different ETLs' effects on the Performance of Proposed double absorber device structures.

Double absorber configuration	J _{sc} (mA cm ⁻²)	V _{oc} (V)	FF (%)	PCE (%)
FTO/STO/CsPbI ₃ /CZTSSe/NiO/W	41.8154	1.021	87.50	37.35
FTO/PCBM/CsPbI ₃ /CZTSSe/NiO/W	36.0730	1.015	87.39	32.01
FTO/C ₆₀ /CsPbI ₃ /CZTSSe/NiO/W	41.5217	1.020	87.31	36.99
FTO/CdS/CsPbI ₃ /CZTSSe/NiO/W	41.7997	1.025	79.92	34.27

efficiency in around 37.35 % due to its good band alignment, lengthy carrier lifetime, and low positive conduction band offset values with the CsPbI₃ absorber layer. Fig. 4 illustrates the J-V curve of Double absorber solar cells equipped with a variety of ETL layers.

3.2. Impact of several HTLs depending on STO as ETL on Double absorber solar cell

Various potential HTL materials are explored here, with STO as the ETL. An effective HTL materials should have the following characteristics: (1) higher carrier mobility for increasing the fill factor; (2) a broader optical energy band gap and higher transparency to alleviate optical loss; (3) higher resistance to water, light, and heat; (4) minimal cost of material and production; (5) friendliness for environmental (Omarova et al., 2022). Since the HTL layer plays a vital role in

**Fig. 4.** The J-V curve of a double absorber solar cell with a variety of ETL materials.

permitting effective hole extraction from the absorber materials towards the back contact, optimizing its structure can significantly boost efficiency. It is investigated how using different HTL materials, such as NiO, CuSCN, spiro-OMeTAD, and Cu₂Te, impacts device performance. Their collected simulation parameters from the literature are shown in Table 5. When contrasting the effectiveness of various HTLs on Double absorber structures, the authors' finding revealed that the NiO-based configuration has the maximum efficiency of all the others and obtained a rating of 37.35 % for its efficiency. This is due to the structure's higher carrier mobility and its lengthy carrier lifetime, as shown in Table 6. Fig. 5 depicts a current–voltage graph for a Double absorber solar cell that is comprised of a variety of different HTL materials.

3.3. Impact of absorber thickness on Double absorber solar cells

The effectiveness of a photovoltaic cell relies heavily on the absorber material and the variations in its thickness. Therefore, generated charge carriers' diffusion lengths must be larger than the solar cell thickness for efficient harvesting. It is best if the photo-generated holes and electrons are balanced by known absorption and recombination (Zhao et al., 2018). Here, absorber layer thickness facilitated charge carrier transport to ETLs and HTLs. Therefore, raising the thickness results in more photon light absorption and enhanced performance of the device (Chakraborty et al., 2019). However, increasing thickness values also enhanced the concentration of the produced charge carrier along the interfaces ETL/CsPbI₃ and CZTSSe/HTL, leading to improvements in recombination rate and, in turn, a decrease in photovoltaic performance such as PCE (%) and V_{oc} (V) (Mehrabian et al., 2021). Here, the bottom absorber materials thickness was changed from 0.5 μm to 2 μm, while all other layer thicknesses were held fixed to investigate the influence on the effectiveness of Double absorber solar cells. The relationship between absorber thickness with V_{oc} and J_{sc} is shown in Fig. 6(a). It is observed in Fig. 6(a) that the V_{oc} was 1.06 V and the J_{sc} was 38.7 mA/cm² when the thickness was 0.5 μm; on the other hand, whenever the thickness was raised to 2.0 μm, the V_{oc} dropped to 1.02 V, and the J_{sc} increased to 42.20 mA/cm², and this improvement because of this strong absorption coefficient. The device's J_{sc} raises as the absorber thickness grows. This is because greater light and carriers are absorbed. This, in turn, causes a sharp decline in the V_{oc}. Fig. 6(b) depicts the FF and PCE curve for various absorber thicknesses, showing that the FF rises with increasing thickness up to around 1.4 μm, after which it declines due to rises in resistance. Fig. 6(b) also demonstrates that PCE is shown to grow before reaching a maximum at a thickness of 1.4 μm, after which it declines significantly. The optimal thickness of a bottom absorber, where efficiency improves, has been proposed to be 1.4 μm.

Table 5

Numerical parameters for each type of HTL material are as follows:

Parameters	NiO (Abedini-Ahangarkola et al., 2022)	Spiro-OMeTAD (Pindolia et al., 2022)	Cu ₂ Te (Suman and Kumar, 2021)	CuSCN (Hima and Lakhdar, 2020)
W (μm)	0.350	0.200	0.08	0.05
E _g (eV)	3.6	2.88	1.18	3.4
x (eV)	1.8	2.05	4.2	1.9
ε _r	11.75	3.0	10.0	9.0
N _c (1/cm ³)	2.5 × 10 ²⁰	2.5 × 10 ²⁰	7.8 × 10 ¹⁷	2.2 × 10 ¹⁸
N _v (1/cm ³)	2.5 × 10 ²⁰	2.5 × 10 ²⁰	1.6 × 10 ¹⁹	1.9 × 10 ¹⁹
V _{te} (cm/s)	1 × 10 ⁷	1 × 10 ⁷	1 × 10 ⁷	1 × 10 ⁷
V _{tp} (cm/s)	1 × 10 ⁷	1 × 10 ⁷	1 × 10 ⁷	1 × 10 ⁷
μ _n (cm ² /Vs)	2 × 10 ⁻¹	2.1 × 10 ⁻³	5 × 10 ²	2 × 10 ⁻⁴
μ _p (cm ² /Vs)	2 × 10 ⁻¹	2.6 × 10 ⁻³	1 × 10 ²	1 × 10 ⁻³
N _D (1/cm ³)	0	0	0	0
N _A (1/cm ³)	5 × 10 ¹⁸	1 × 10 ¹⁸	1 × 10 ²¹	1 × 10 ¹⁸
N _t (1/cm ³)	1 × 10 ¹⁵	1 × 10 ¹⁵	1 × 10 ¹⁵	1 × 10 ¹⁵

Table 6

Performance Evaluation of the proposed Double absorber Structure Using distinct HTL.

Double absorber Structure	V _{oc} (V)	J _{sc} (mA/cm ²)	FF (%)	PCE (%)
FTO/STO/CsPbI ₃ /CZTSSe/NiO/W	1.021	41.81543	87.50	37.35
FTO/STO/CsPbI ₃ /CZTSSe/Cu ₂ Te/W	0.890	41.760	86.91	32.31
FTO/STO/CsPbI ₃ /CZTSSe/Spiro-OMeTAD/W	1.016	41.81198	86.10	36.57
FTO/STO/CsPbI ₃ /CZTSSe/CuSCN/W	1.021	41.81542	87.49	37.34

3.4. The impact of bulk defect concentration on the functional performance of a proposed double absorber solar cell

The defect is crucial to the device's effectiveness. When more defects per unit area are in the absorber layer, more pin holes are created, the film degrades more quickly, and the device's stability and performance suffer (Jamal et al., 2019). Defects are classified as deep, or shallow based on their depth and location. Shallow defects have a defect density of around 1 × 10¹⁰–1 × 10¹³ cm⁻³, whereas deep defects have a defect concentration of about 1 × 10¹⁴ cm⁻³ and go up to about 1 × 10¹⁶ cm⁻³ in size (Adinolfi et al., 2016). The presence of deep defects characterizes the central band gap. The absorber layer is well-known to be the primary site of photoelectron production and the location of recombination centres, i.e., Shockley-Read Hall (SRH), because of bulk defects with energy levels deeper in the band gaps. When the absorber material defect concentration is equivalent to or more than the doping concentrations of the absorber material, the device will become semi-insulating, and it will be impossible to produce an appropriate p-n junction. As a result, the device fails to be operating correctly, and its overall performance degrades. To explore defect densities impact in each layer, the SRH model was utilized, which is expressed in Eqns. (8) and (9).

$$R_{SRH} = \frac{np - n_i^2}{\tau_p(n + n_i) + \tau_n(p + p_i)} \quad (8)$$

$$\tau_{n,p} = \frac{1}{\sigma v_{th,n,p} N_t} \quad (9)$$

Here The impact of varying the trap concentration of the bottom absorber layer, CZTSSe, is explored from 1 × 10¹⁰ to 1 × 10¹⁹ while holding the trap concentration of the other absorber layers kept constant at 1 × 10¹⁴ because, after the simulation, it is found that top absorber defect has a less significant effect on device performance. In Fig. 7, the impact of the cell's J_{sc}, V_{oc}, PCE, and FF with respect to bulk defect is seen. At the time of altering the trap density in CZTSSe, the J_{sc} and V_{oc} of a Double absorber device decline due to greater SRH recombination and a lower carrier lifetime as trap density rises. In Fig. 7(a), it is shown that J_{sc} and V_{oc} of the cell fall from 41.8 mA/cm² to 3.81 mA/cm² and 1.24 V to 0.67 V accordingly, as the defect concentration of CZTSSe increases, whereas PCE and FF decline from 43.3 % to 1.29 % and 83.3 % to 51.2 % correspondingly shown in Fig. 7(b). Nonetheless, FF has been on the rise once more once the defect concentration reached 1E18. To get the best results, we settle on a bulk defect concentration of 1 × 10¹² cm⁻³, corresponding to an optimal J_{sc} of 41.82 mA/cm², V_{oc} of 1.0207 V, PCE of 37.35%, and FF of 87.50 %.

3.5. Double absorber solar cell performance under different temperature conditions

Working (i.e., operational) temperature significantly influences device performance, which changes with environmental situations. Most of the simulations here employed an operating temperature (ambient temperature) of 300 K. Moreover, if the solar panels are mounted outside, the sun's rays will heat them, causing an increase in the working

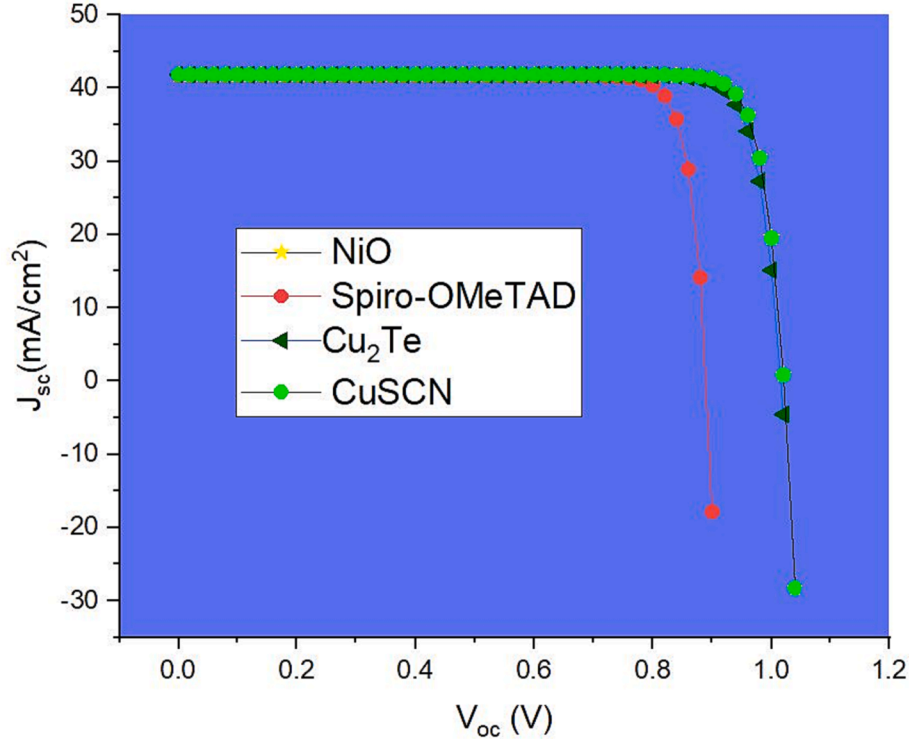


Fig. 5. Current-voltage curve of double absorber solar cells developed with a variety of HTL materials.

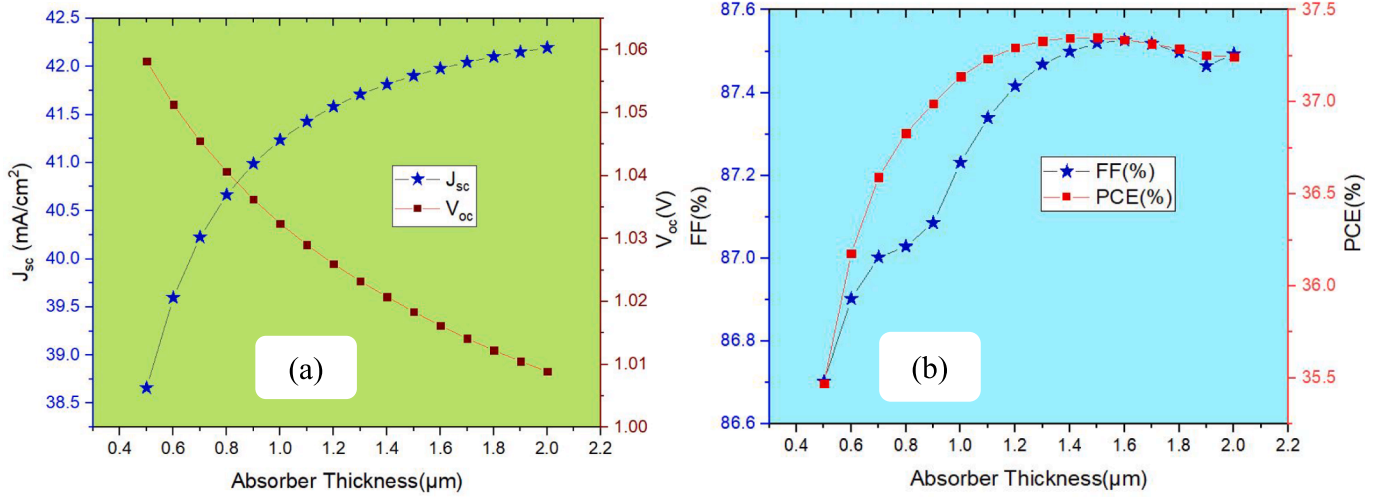


Fig. 6. Influence of Bottom Absorber thickness on (a) current-voltage curve, (b) Fill Factor, and PCE.

temperature. Because of this, the photovoltaic cells often operate at temperatures greater than 300 K. The change in operating temperature was done between 260 K and 420 K. At the same time, the illumination level is held constant at 1000 W/m² for investigating the impact of that working temperature on cell performance. Fig. 8(a) and (b), plotting several device output characteristics against temperature, demonstrate that increasing the working temperature consistently reduces cell performance. As seen in Fig. 8 (a), As temperature rises, there is a considerable decline in V_{oc} , which goes from 1.04 V to 0.94 V. However, when the temperature of the material is elevated, the materials band gap will narrow. This will result in a greater number of electron-hole pairs being created, increasing the value of J_{sc} . In reality, reverse saturation current density (J_0) rises with rising temperature, which corresponds to a decline in V_{oc} according to equation (10) (Alam and Ashraf, 2020).

$$V_{oc} = \frac{nkT}{q} \log \left(\frac{J_{sc}}{J_0} + 1 \right) \quad (10)$$

Where J_0 = reverse saturation current, J_{sc} = current density, V_{oc} = open circuit voltage, K = Boltzmann constant, q = electronic charge, T = temperature, and n = ideality factor. Fig. 8(b) demonstrates that when the working temperature of the cell is raised, both the PCE and FF values will drop. This decline can be attributable to an increase in SRH recombination, poorer mobility, the enhanced series resistance of the cell at high temperatures, as well as a shorter carrier diffusion length (Coulibaly et al., 2019). Additionally, there is a linear correlation between the operating temperature and device efficiency, so lower temperatures result in greater device efficiency.

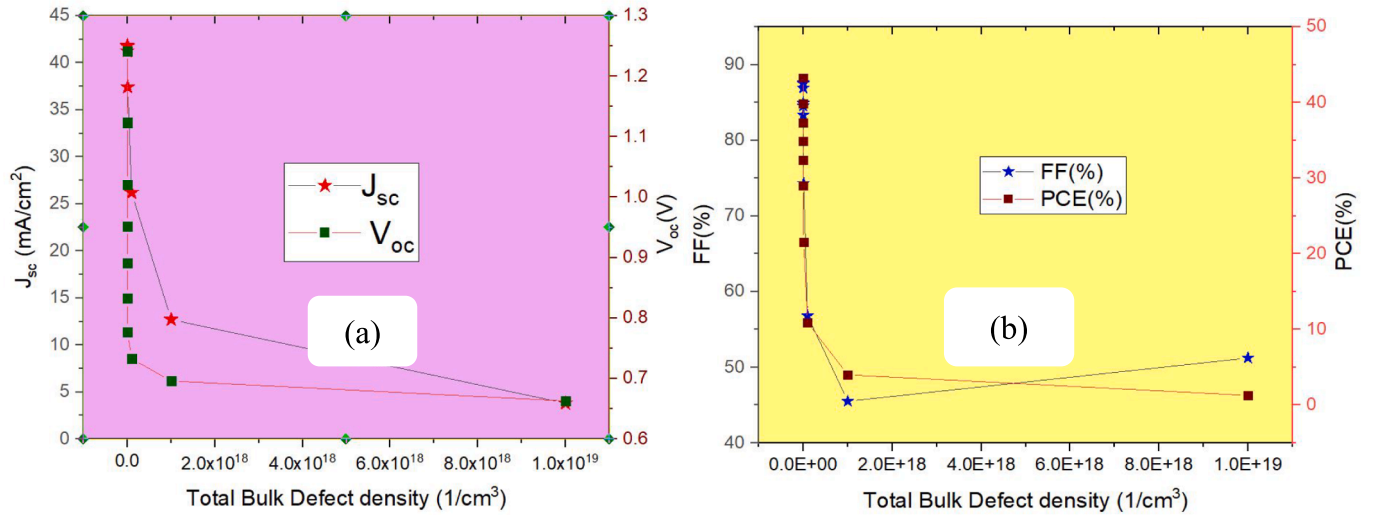


Fig. 7. Influence of Bottom Absorber Layer Bulk Defect on (a) J-V Curve (b) Fill Factor and PCE.

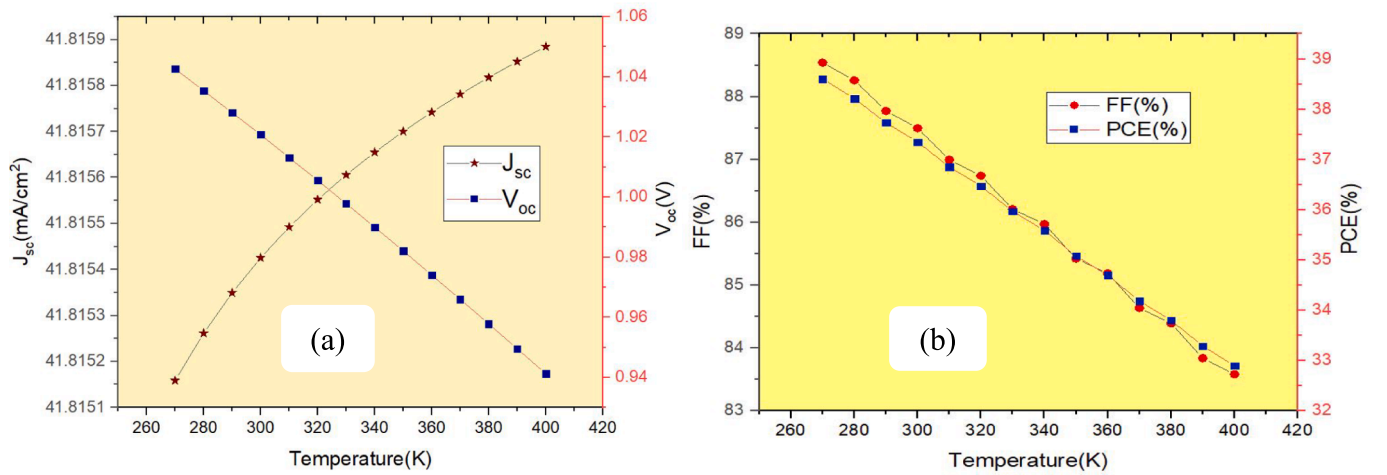


Fig. 8. Impact of working Temperature on the proposed device: (a) J_{sc} and V_{oc} (b) PCE and FF.

3.6. Impact of diverse back contact materials on the effectiveness of double absorber solar cells

The back contact work function is essential for achieving an acceptable built-in voltage (V_{bi}) in photovoltaic cells (Raoui et al., 2019). The work function of various materials creates variable series resistance with its cell structure, which affects the cell's performance. Numerous materials with varying work functions, including carbon (5.0 eV), gold (5.1 eV), tungsten (5.2 eV), Palladium (5.3 eV), nickel (5.5 eV), platinum (5.65 eV), and selenium (5.9 eV), are investigated to figure out the influence of work function of the metal electrode on photovoltaic characteristics (Tariq Jan and Noman, 2022; Omarova et al., 2022; Deepthi Jayan and Sebastian, 2021; Raoui et al., 2019). Energy band graphs for various back contacts with the absorber are presented in Fig. 9. The outcomes of simulation with respect to change in anode material are represented in Fig. 10. Whenever the work function is raised from 5 eV to 5.22 eV, the results reveal that the J_{sc} , FF, and PCE all rise by a significant amount. This happens because the Schottky barrier between the HTM and the back contact is lowered, making it simple for holes to go from the HTM to the back contact. The maximum efficiency of a cell is reached at 5.22 eV. Therefore, it is recommended that the anode electrode be made of base metal with a greater than 5.22 eV, such as W, Pd, Ni, Pt, or Se. Since tungsten (as seen in Fig. 1(a))

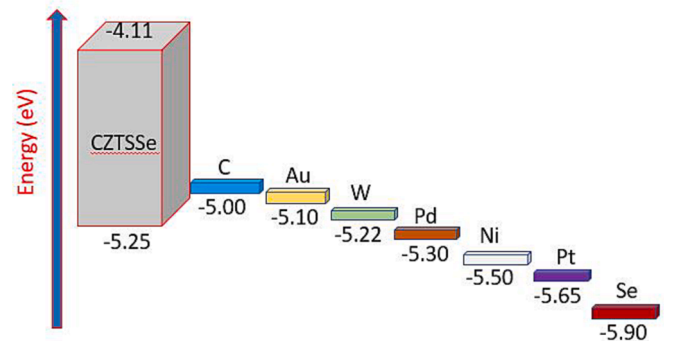


Fig. 9. Band Alignment of the CZTSSe absorber with a variety of back contact materials.

is inexpensive and has a desirable work function, we choose this material in our simulation model as back contact. The results of a comparison between the PV performance of the present study and those of other previously published studies are listed in Table 7.

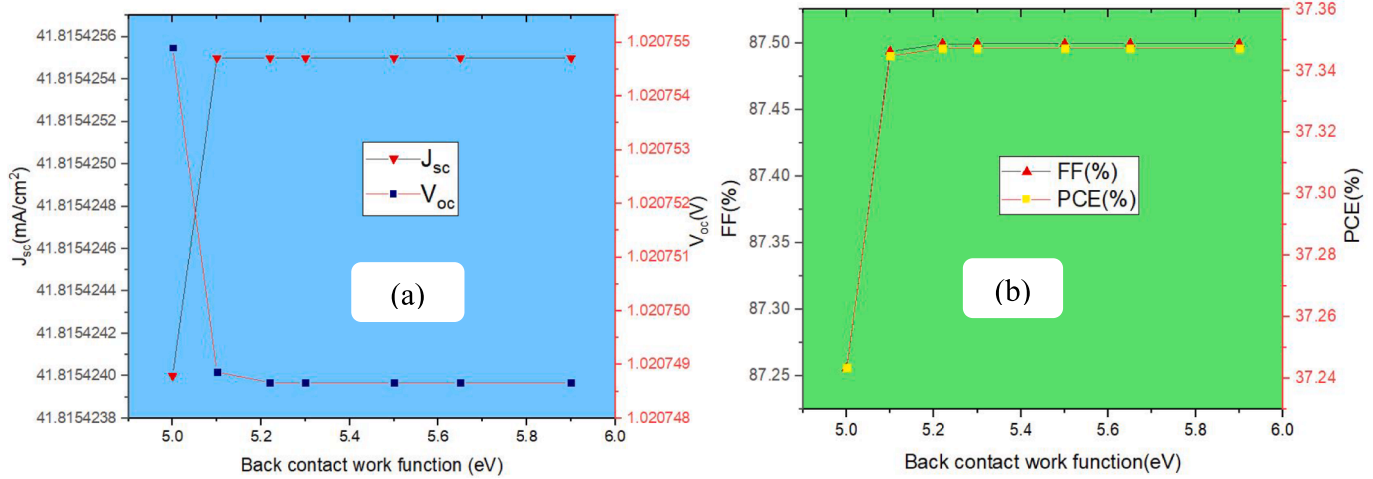


Fig. 10. Influence of Back contact Materials on the device: (a) V_{oc} and J_{sc} (b) FF and PCE.

Table 7

Comparison of PV parameters of present work with other Double absorber structures from the published literature.

Double absorber Structure	J_{sc} (mA/cm ²)	V_{oc} (V)	FF (%)	PCE (%)	Reference
CZTSe/CZTS (Simulation)	25.83	0.98	69.87	17.81	(Enayati Maklavani and Mohammadnejad, 2020)
CZTS/Silicon (Simulation)	27.55	0.84	88.54	19.40	(AlZoubi et al., 2021)
MASnI ₃ /MAPbI ₃ (Simulation)	30.87	1.15	85.29	30.29	(Abedini-Ahangarkola et al., 2022)
NaZn _{0.7} Cu _{0.3} Br ₃ /MASnI ₃ (Simulation)	34.98	1.0054	87.78	30.87	(Cheragee and Alam, 2022)
NaZn _{0.7} Cu _{0.3} Br ₃ /MASnI ₃ (Simulation)	35.03	1.0054	87.79	30.92	(Cheragee et al., 2022)
NaZn _{0.7} Cu _{0.3} Br ₃ /MASnI ₃ (Simulation)	34.71	1.1373	82.13	32.42	(Cheragee and Alam, 2023)
CsPbI ₃ /CZTSSe (Simulation)	41.8154	1.0207	87.50	37.35	Present Work

4. Conclusion

This research uses the SCAPS-1D software program to conduct a numerical analysis and optimization of a Double absorber solar cell structure. First, it is looked at different organic–inorganic compounds that might be used for HTL and ETL. The simulation findings suggest that STO works well as an ETL and NiO works well as an HTL because of their strong band alignment with absorbers and increased mobility, respectively. Besides, the Effects of back contact work function, absorber thickness, operating temperature, and defect density of the absorber are also investigated. Based on the simulation results, metals with a work function larger than 5.20 eV are good candidates for this double absorber structure's back electrode. It is shown that a bottom absorber defect density is not more than $1 \times 10^{12} \text{ cm}^{-3}$ is necessary for optimal device performance and that the efficiency of the device steadily declines with a rise in operating temperature and absorber layers defect density of the cell. Power conversion efficiency optimization at 300 K reveals that a thickness of 0.01 μm for a CsPbI₃-based absorber layer and 1.4 μm for a CZTSSe-based absorber layer yields a maximum efficiency of 37.35 %, and this information will be helpful for both future research and the Experimental Framework.

CRediT authorship contribution statement

Sheikh Hasib Cheragee: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Mohammad Jahangir Alam:** Investigation, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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